Deep learning Project Presentation

ParkVision

Parking Space Detection

Automated occupancy classification using YOLOv8n
trained on the PKLot dataset across 3 parking environments

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CONTEXT

Problem Statement

01

Crowded Parking

Finding available parking in dense urban areas is time-consuming and stressful for drivers.

02

Manual Monitoring

Labor-intensive, error-prone surveillance cannot scale with growing city infrastructure demands.

03

Automated Solution

Vision-based deep learning using existing camera infrastructure provides a scalable alternative.

Task: classify each parking slot as empty or occupied · Challenge: generalize to unseen parking environments (cross-domain)

DATA

Dataset — PKLot

3 Parking Lots

PUCPR · UFPR04 · UFPR05

3 weather conditions each: Sunny / Cloudy / Rainy

Image Resolution

1280×720 → resized to 640×640

Annotations converted: XML → YOLO format

LOT-BASED TRAIN / VAL / TEST SPLIT

Train: PUCPR + UFPR04

Validation: UFPR05 — first half

Test: UFPR05 — second half

Entirely unseen during training → direct measure of cross-domain generalization

Training instances: ~289,619 empty · ~238,351 occupied · Balanced across splits

MODEL

Methodology

ARCHITECTURE

YOLOv8n

Nano variant
Lightweight — suitable for resource-constrained deployment
Anchor-free detection head
Decouples classification & regression
Compact size
Reduces overfitting risk across domains

TRAINING SETUP

Google Colab
Batch 16 · Image 640×640 · Ultralytics lib

EVALUATION METRICS

mAP50	mAP50-95
Precision	Recall

3-iteration evidence-based development: each iteration identifies weaknesses from prior model, proposes targeted fix, validates improvement on test set

ITERATION 1

Baseline — No Augmentation

CONFIGURATION

augment=False
Default YOLOv8n config
15 epochs

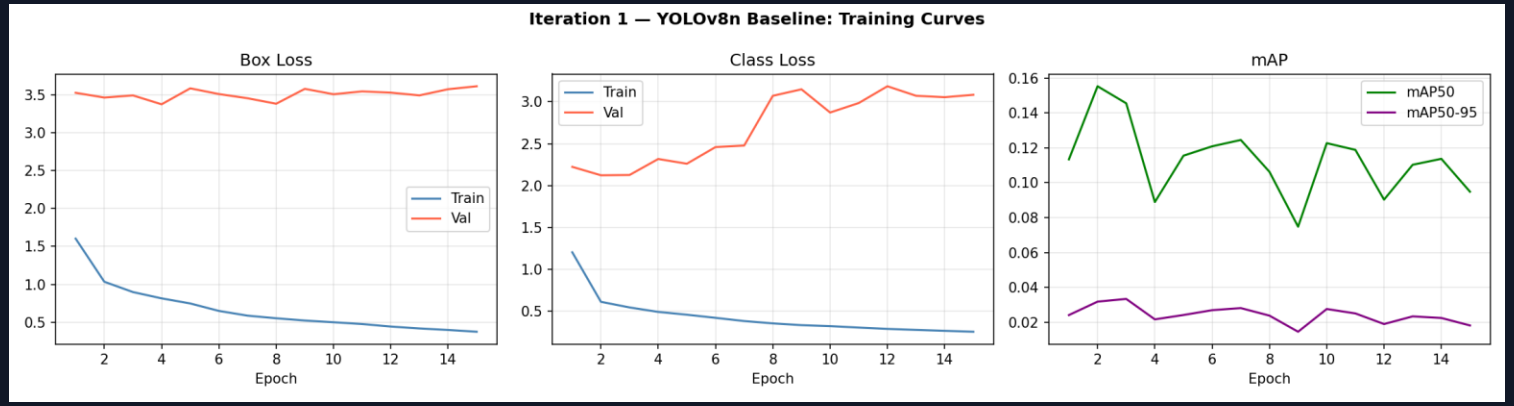
OVERFITTING

TEST SET RESULTS

mAP50	mAP50-95
0.1472	0.0342
Precision	**Recall**
0.2934	0.2437

ITERATION 1

Learning Curves



Finding: Train-val loss gap widened continuously across all 15 epochs. Without augmentation the model memorizes training-set weather & camera conditions and fails to generalize to unseen parking lots.

ITERATION 2

+ Augmentation & Regularization

CHANGES FROM ITER 1

- augment=True -> expose model to lighting and viewpoint variation
- dropout=0.2, weight_decay=0.0005 -> penalize memorization
- lr0=0.005, warmup_epochs=5 -> smoother convergence
- box=7.5 -> emphasize localization

extended to 50 epochs with patience=20.

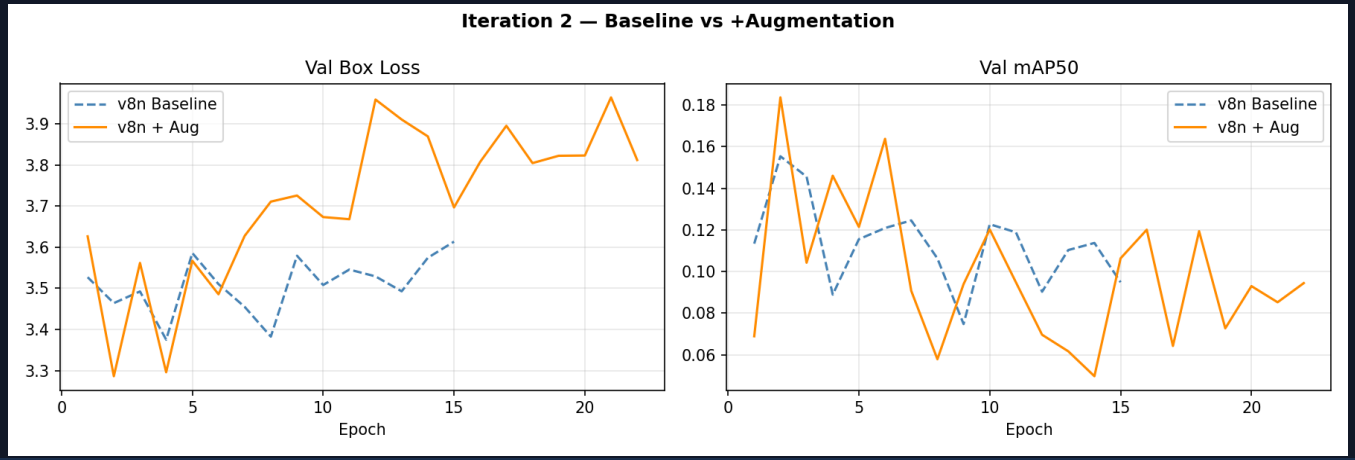
UNSTABLE LEARNING

TEST SET RESULTS

mAP50	mAP50-95
0.1864	0.0417
Precision	**Recall**
0.3325	0.3209

ITERATION 2

Learning Curves



Finding: val mAP50 peaked early (~epoch 2) then collapsed. lr0=0.005 and full Mosaic (prob=1.0) destabilize training. Augmentation alone is insufficient, learning rate is too high.

ITERATION 3 ★ FINAL

+ Augmentation (Tuned)

CHANGES FROM ITER 2

- lr0=0.003 (↓ from 0.005) -> smaller gradient steps
- mosaic=0.5 (↓ from 1.0) -> halved probability
- warmup_epochs=8 (↑) -> longer warm-up
- dropout=0.1 (↓) -> reduced regularization to avoid underfitting

Same 50 epochs with increase patience=30 -> larger recovery budget

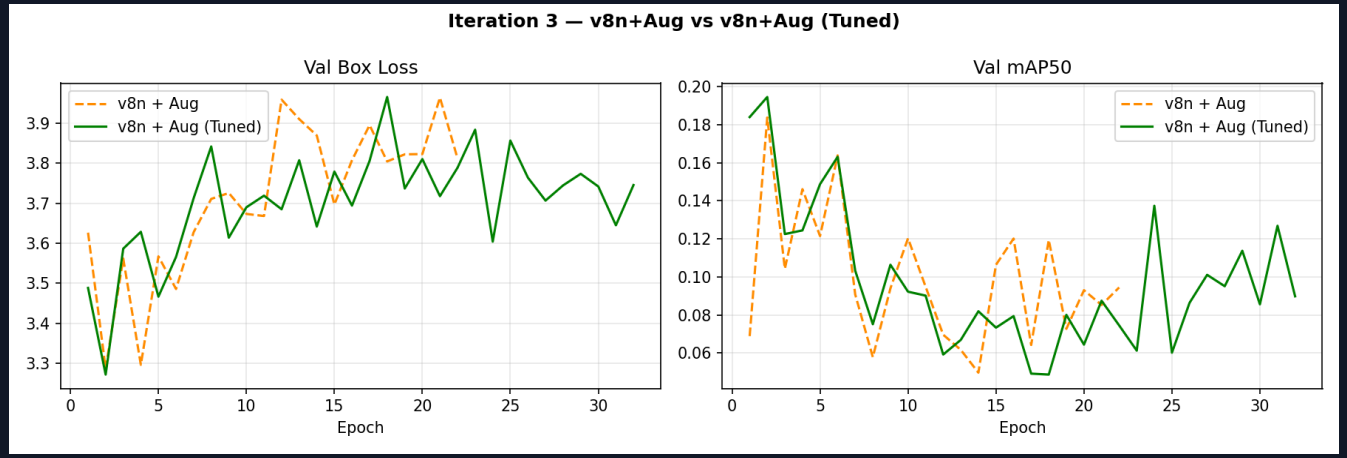
Metrics Score Improved

TEST SET RESULTS

mAP50	mAP50-95
0.1968	0.0456
Precision	**Recall**
0.3150	0.3213

ITERATION 3 ★ FINAL

Learning Curves

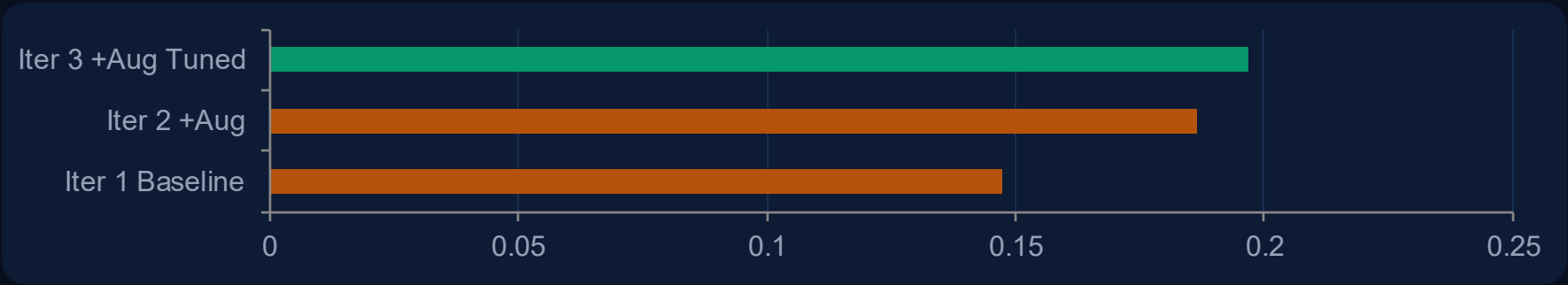


Result: val mAP50 peaked higher (0.19 vs 0.18) and remained stable — no post-peak collapse. Reducing lr0 and Mosaic probability stabilizes training. Best performance across all 4 metrics.

RESULTS

Model Comparison

Model	mAP50	mAP50-95	Precision	Recall
1. YOLOv8n Baseline	0.1472	0.0342	0.2934	0.2437
2. + Augmentation	0.1864	0.0417	0.3325	0.3209
3. + Aug Tuned ★	**0.1968**	**0.0456**	**0.3150**	**0.3213**



ANALYSIS

Challenges & Limitations

Cross-Domain Shift

Test lot (UFPR05) has different layout & camera angles never seen during training. Regularization cannot bridge this gap — it is not classical overfitting.

Limited Diversity

Only 2 parking lots in training. More varied environments including UFPR05-like conditions in training would reduce domain shift.

DEPLOYMENT

- **Docker** — containerize model + dependencies
- **Streamlit** — web-based demo interface for live inference

CONCLUSION

Summary & Future Work

3-iteration process: Each iteration motivated by specific metric weaknesses from the prior run — evidence-based development

mAP50 = 0.197: +33.8% improvement over no-augmentation baseline · mAP50-95 improved +33.3%

Augmentation balance: mosaic=1.0 destabilizes training; mosaic=0.5 with lr0=0.003 is the optimal configuration

Domain shift: Remaining performance gap is cross-domain generalization failure — not classical overfitting

FUTURE WORK

1. Domain adaptation techniques to reduce cross-lot feature divergence
2. Multi-lot training strategies — include UFPR05-like conditions in training set